

Thinking in Systems

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THE ARGUMENT

Why do the problems we work hardest on refuse to go away?

Because the effort lands on events, and the behaviour lives in the structure. Hunger, addiction, overfishing, arms races, traffic and unemployment persist through decades of analytical brilliance because each one is the characteristic output of an arrangement of stocks, flows, feedback loops and goals that nobody redesigned. Swap the elements and a system stays itself. Change the interconnections and it becomes something else. Change its purpose and it becomes something else again. The behaviour is latent in the structure the way the bounce is latent in a Slinky, and it will keep reappearing, under new names and new leadership, until the structure changes. That is the unsettling half of systems thinking. The comforting half is that structure is something people build, which means it is something people can rebuild.

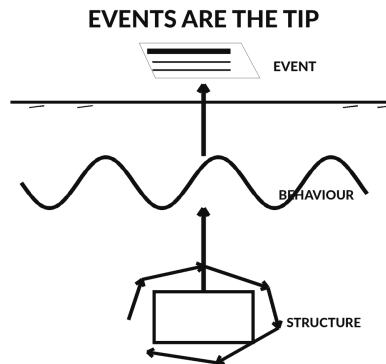
THE TWELVE LEVERAGE POINTS

A list of places to intervene in a system, ordered from the weakest to the strongest. It was written in one sitting out of frustration during a meeting on global trade, and it is offered as an invitation to think more broadly rather than as a recipe: the order is slithery and every item has exceptions that move it. One rule holds across the whole list. The higher the leverage point, the harder the system will resist changing it.

- 12 **Numbers** Constants and parameters — subsidies, taxes, standards, the minimum wage, the size of a flow. Almost all attention goes here and almost no leverage is here.
- 11 **Buffers** The size of a stabilising stock relative to its flows. Rivers flood, lakes rarely do. Buffers steady a system, but they are usually physical and costly to change.
- 10 **Stock-and-flow structure** The plumbing itself. Powerful, and rarely a leverage point, because rebuilding is slow and expensive. The leverage is in laying it out correctly the first time.
- 9 **Delays** Lengths of time relative to the rate of change. Delays determine how fast a system can react and how far it overshoots — and mostly they cannot be shortened. Things take as long as they take.
- 8 **Balancing feedback loops** The strength of a correcting loop relative to the impact it has to correct. Stripping out loops that are rarely used, because they look like cost, quietly narrows the range a system can survive.
- 7 **Reinforcing feedback loops** The gain around a driving loop. Slowing the growth is almost always a stronger move than strengthening the brakes that are trying to contain it.
- 6 **Information flows** Who does and does not have access. Adding a loop where none existed is the cheapest powerful intervention available, and the one most reliably resisted by whoever benefits from its absence.

- 5 **Rules** Incentives, punishments, constraints — the scope and degrees of freedom of the whole system. Power over the rules is real power. Watch who holds it and what they are allowed to exclude.
- 4 **Self-organisation** The power to add, change or evolve structure. The strongest form of resilience, and it requires variability, experimentation and a willingness to lose control, which is why it is so often suppressed.
- 3 **Goals** The purpose of the system. Everything lower on this list — structure, loops, information, even self-organisation — bends to conform to it.
- 2 **Paradigms** The shared unstated assumptions the goals and the structure arise from. Societies resist paradigm change harder than they resist anything else. In a single person it can happen in a moment.
- 1 **Transcending paradigms** Staying unattached to any of them; holding that no paradigm is true, including the one shaping your own view, and finding that funny rather than frightening.

INSIGHTS

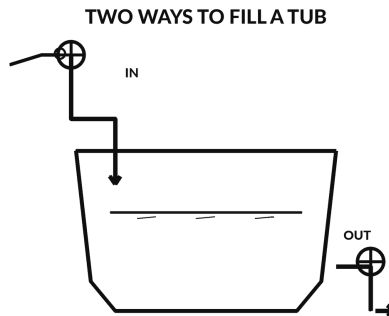


The bounce is in the Slinky

A system's characteristic behaviour is latent inside its own structure. An outside event can release that behaviour or suppress it, but the same event applied to a different system produces a different result. The system, to a large extent, causes what it does.

Meadows opened her classes by balancing a Slinky on one upturned palm, gripping it from above, and pulling the lower hand away. The spring drops, springs back, and yo-yos in the air. Then she does exactly the same thing with the cardboard box the Slinky came in. It hangs there. Same hand, same release, different object, different behaviour.

- What follows is a ranking. Events are the most visible layer of a system and the least explanatory — which is why nightly market commentary predicts nothing. Underneath sits behaviour, the shape a variable makes over time. Underneath that sits structure: interlocking stocks, flows and loops. Ask for the time graph before you accept the explanation.



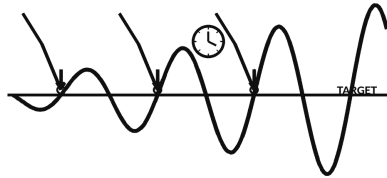
There is more than one way to fill a bathtub

The mind fastens on stocks more readily than on flows, and on inflows more readily than on outflows. So half the available levers go unnoticed. A stock rises when you increase what comes in, and it rises just as surely when you reduce what goes out.

A breakthrough in energy efficiency does the same thing to the stock of available oil as the discovery of a new field. Different people profit from it, which is most of why one feels like an achievement and the other feels like a sacrifice. A workforce grows by hiring or by fewer people leaving; national wealth rises by building factories or by making them last longer, often more cheaply.

- Stocks change slowly because flows take time to flow, and that slowness is the pace-setter of the whole system. It also buys room to manoeuvre: soil that took centuries to accumulate rarely erodes at once, and skills a population has learned are not forgotten overnight. Expect momentum, plan for it, and stop giving up too soon.

STEER FASTER, SWING WIDER



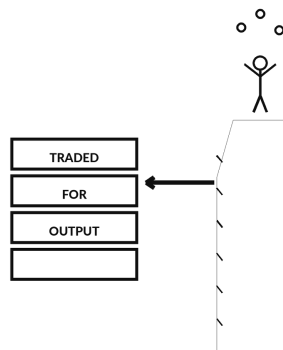
Reacting faster widens the swing

A delay inside a balancing loop makes a system oscillate. The oscillation is not evidence that anyone involved is slow or careless. It is what the structure does, and the people inside it are struggling to steer on information that cannot arrive in time.

A car dealer averages sales over five days, corrects a third of any shortfall per order, and waits five days for delivery. One permanent ten per cent rise in demand sets her inventory swinging for months. Shortening her perception delay makes the swings slightly worse. Cutting her response time from three days to two makes them far worse. Lengthening it to six damps them and the system settles.

- This is the signature failure of intervention: high leverage, wrong direction. She found a real lever and pushed it the wrong way. The same structure spread across an industry — perception, production, delivery and construction delays entrained with each other and amplified by speculators — is the primary cause of business cycles. Presidents can sharpen or soften them. They do not create them.

THE LEDGE GETS NARROWER



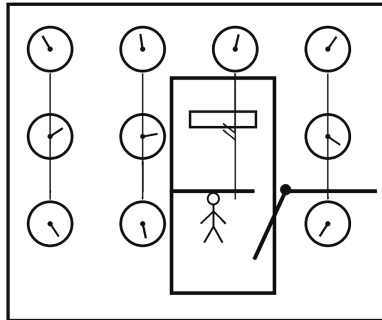
Resilience is spent invisibly

Resilience is the plateau a system can play on — the range it can wander over and still be returned by its own restoring loops. Stability is what you can measure week to week. Resilience is what you cannot see at all until you exceed it.

Because it is invisible it gets traded for things that are not. Meadows attributes to bovine growth hormone a rise in milk yield achieved by diverting the cow's metabolic energy from other functions, leaving the animal less healthy, shorter-lived and more dependent on human management. Just-in-time delivery cut inventory costs across whole industries and made production vulnerable to any interruption in fuel, traffic or labour.

- The plateau narrows quietly, the walls get lower, and one day the system does something it has done a hundred times before and crashes. Racing sailboats are the pure case: optimised so completely around the current rules that any change to the rules would render them useless. Manage for the ability to recover, not only for output and steadiness.

ONE GAUGE, WHOLE PANEL



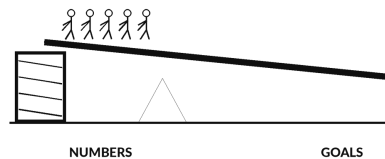
Everyone in the booth sees one gauge

People make quite reasonable decisions on the information reaching them, and the information reaching them is never information about the whole system. Herbert Simon called this bounded rationality: not omniscient optimisers, but satisficers finding something they can live with for now and sticking to it.

Fishermen do not know how many fish there are, much less how many will be caught by everyone else today. Investors do not know what other investors are planning, so they over-invest and under-invest together. The finding is not that these people are foolish. It is that anyone placed there decides the same way. Students playing simulated fishermen overfish. Given a manager's position, you will start seeing labour as a cost.

- The practical consequence is unwelcome and freeing at once: replacing the individual almost never changes the outcome, and blaming them almost never helps. What changes the outcome is redesigning what reaches that position — the information, the incentives, the goals, the constraints. Understanding why the behaviour is reasonable is not an excuse for it. It is the only route to changing it.

THE CROWD PUSHES THE SHORT END



The crowd pushes the short end of the lever

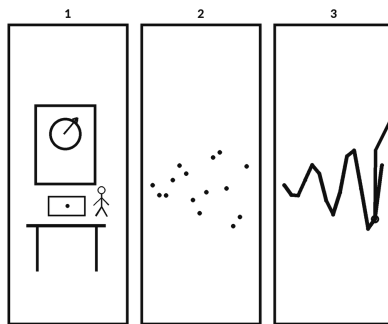
Leverage points are places where a small change produces a large shift in behaviour. They exist, people deep inside a system often sense where they are, and more often than not they push them backwards — systematically worsening the problem they set out to solve.

The world model Forrester built for the Club of Rome returned growth itself, in population and in the economy, as the leverage point. World leaders were fixated on exactly that lever and pushing it hard the wrong way. His 1969 urban dynamics study found subsidised low-income housing to be a leverage point in the direction nobody expected; he was derided at the time, and many of those projects have since been torn down.

- Ranked weakest to strongest the list runs numbers, buffers, physical structure, delays, balancing loops, reinforcing loops, information flows, rules, self-organisation, goals, paradigms. Meadows' own estimate was that some ninety-nine per cent of attention goes to the first item. Parameters matter enormously to whoever is standing in the flow, and changing them rarely changes what the system does.

ACTIONS

PLOT IT BEFORE YOU FIX IT

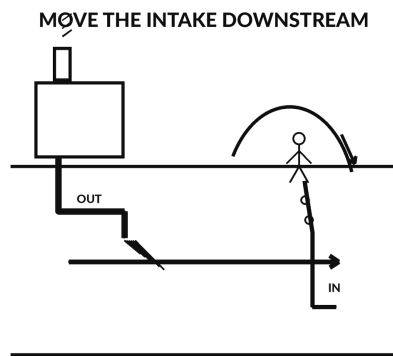


Plot the history before you propose anything

Before disturbing a system in any way, watch how it behaves. Learn its history. Ask the people who have been around a long time what actually happened. Where data exist, plot them over time rather than relying on anyone's memory, including your own.

- Use this at the start of any diagnosis, and above all when everyone in the room already agrees on the cause. It has worked when the question changes from what is wrong to how did we get here, what else could this do, and where does this end if nothing changes. Plotting several variables together starts to show not only what is in the system but how it might be wired.

Memory is unreliable about timing and direction, and confident people are no exception. Meadows records being told with great authority that milk prices were rising when they were falling. Starting from behaviour also blocks the commonest move in any meeting, which is to define the problem as the absence of one's favourite solution.



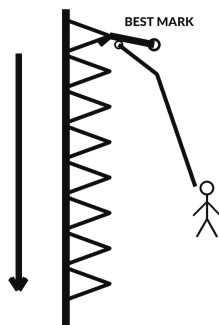
Move the intake pipe downstream of the outflow

Find the decision being made without feedback about its own consequences, and route that feedback back to whoever is making it, in a form they cannot route around. Missing information flows are among the most common causes of system malfunction and among the cheapest to repair.

- Use it wherever the cost of a decision lands somewhere other than on the decider — a shared resource, a discharge, a budget, a rule its authors exempted themselves from. Success looks like behaviour changing with no penalty attached. After the Toxic Release Inventory made United States factory emissions a matter of public record in 1988, reported emissions nationwide fell by around forty per cent within two years. No fines, no lawsuits, no required reductions.

Placement and form decide whether it works. Telling every user of an aquifer that the water table is falling can start a race to the bottom; making the price of water climb steeply as pumping outruns recharge does not. Expect resistance in proportion to the value of the missing loop, because someone is usually benefiting from its absence.

SET THE CATCH AT YOUR BEST MARK



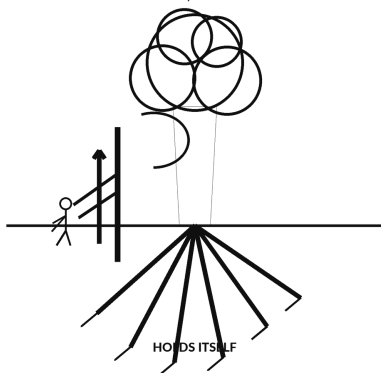
Set the catch at your best mark, not your last one

Hold the performance standard absolute. Where a standard has to move at all, let it be pulled upward by the best result you have ever achieved rather than dragged downward by the worst — which is the direction it will drift on its own.

- Use it wherever a goal is quietly allowed to follow perceived performance: service quality, safety records, a training programme, a country's sense of what it can expect of itself. It is working when the same structure runs in reverse — the better things get, the harder the effort to improve them, a reinforcing loop climbing instead of sinking.

The failure mode is gradual by design, which is why it has a nickname about frogs and warming water. A system that drops sharply provokes correction; one that sinks slowly erases the memory of how good things used to be, and each lowered expectation licenses the next. The tell is bad results treated as the real signal and good ones dismissed as flukes.

THICKEN THE TRUNK, THEN PULL THE STAKE



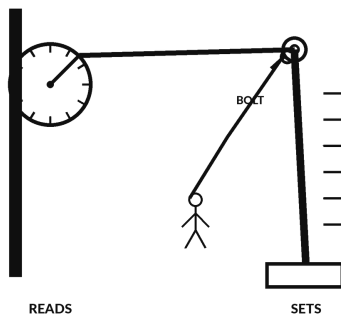
Thicken the trunk, then pull the stake

When you step in to help a struggling system, intervene in the way that rebuilds its own capacity to hold itself up, and then take yourself out. Helping a system to help itself is usually cheaper and easier than taking it over and running it.

- Use it any time a fix relieves a symptom without touching what produced it: a subsidy, a pesticide, a price cap, a painkiller. Start with questions rather than a takeover — why is the system's own correction failing, what is obstructing it, how could it be made to work? The measure of success is needing less of the intervention over time rather than more.

If the intervention lets the original capacity atrophy, a reinforcing loop starts and the system needs ever more of the thing that is weakening it. Withdrawal then means facing the real and by now worse state of things. Sometimes it can be tapered: insulation and efficient cars ahead of a fuel price rise, polyculture and rotation ahead of dropping a pesticide. Sometimes there is no way through but to bear it.

BOLT THE DIAL TO THE LEVER



Bolt the dial to the lever

Write policies that read the state of the system and adjust themselves, instead of policies that fix a number and hope the world holds still. The best of them contain loops that alter and correct their own loops — learning designed into the management process.

- Use it wherever there is real uncertainty about the size of a problem or how fast it will move. The 1987 Montreal Protocol set phase-out targets for ozone-damaging chemicals and also required monitoring and a reconvened congress to revise the schedule — which happened three years later, accelerated, because the damage was worse than foreseen. Success looks like a policy that never needs an emergency rewrite.

A self-adjusting policy is only as good as the indicator it reads, and a system will produce exactly what you ask it for. An indicator that measures effort returns effort: money spent per student, or the gross national product, which rises when there are more car accidents and repair bills and falls when a light bulb lasts ten times as long.

VOCABULARY

Systems language is largely a translation of things people already know intuitively into terms precise enough to reason with. A society that has no working word for a thing does not manage that thing — which is why the words below are worth the small effort of learning.

Stock

An accumulation you can see, count or measure at a given moment: water in a tub, money in an account, wood in a forest, a reserve of goodwill. A stock is the present memory of the flows that have changed it.

Flow

What fills or drains a stock — births and deaths, purchases and sales, hiring and quitting. Flows can be changed abruptly; the stocks they feed cannot.

Balancing feedback loop

A goal-seeking loop that opposes whatever direction of change is imposed on it. A source of stability, and equally a source of resistance to change.

Reinforcing feedback loop

A self-multiplying loop: the more there is, the more is added. It produces exponential growth or runaway collapse, and it is found wherever something can reproduce itself out of itself.

Shifting dominance

When competing loops change relative strength and one takes over, turning growth into decline, or a steady state into a swing. Most complex behaviour is dominance changing hands.

Delay

The time between an action and its effect. Every stock is a delay. A delay inside a balancing loop is what makes a system oscillate.

Nonlinearity

A relationship in which cause does not produce proportional effect. Two hundred pounds of fertiliser does not double the yield from one hundred, and three hundred lowers it.

Limiting factor

At any moment the scarcest input is the one that governs. More of an abundant input does nothing — and growth itself keeps changing which input is scarce.

Bounded rationality

People decide sensibly on the information that reaches them, which is never information about the whole system. Not omniscient optimisers; satisficers, meeting a need well enough and moving on.

Resilience

The ability to recover from disturbance — the range a system can wander over and still be pulled back. Not the same as stability: a system can look steady for years and have almost none.

Leverage point

A place where a small change produces a large shift in behaviour. Usually counterintuitive, frequently identified correctly, and then pushed in the wrong direction.

QUICK REFERENCE

THREE LEVELS	Event, behaviour, structure. Events explain nothing. Ask for the time graph before you accept the story.
TWO LOOPS	A balancing loop seeks a goal and resists change. A reinforcing loop multiplies whatever direction it is already going.
STOCKS	Slow, buffering, pace-setting. Raise one by closing the drain as readily as by opening the tap.
DELAYS	Every stock is a delay. A delay inside a balancing loop means oscillation, and correcting faster widens the swing.
LIMITS	The scarcest input governs. Growth changes which input is scarce, so the binding limit keeps moving.
THE TRAPS	Policy resistance. Tragedy of the commons. Drift to low performance. Escalation. Success to the successful. Addiction. Rule beating. The wrong goal.
LEVERAGE, WEAK TO STRONG	Numbers, buffers, structure, delays, balancing loops, reinforcing loops, information, rules, self-organisation, goals, paradigms.
BEFORE INTERVENING	Get the beat. Put your model where it can be challenged. Send the consequence back to whoever decides.
CHECKING A GOAL	A system produces what you ask it for. If the indicator measures effort, effort is what you will get.

You cannot control a system and you cannot fully figure one out. You can watch it, redesign it, and dance with it.